

ESAT Chemistry Formula + Things to Remember

A compact, calculation-heavy sheet for equations, units, reaction patterns, tests and fast no-calculator chemistry reasoning.

ESAT Chemistry style: expect mole ratios, bonding/properties, rates, energetics, electrolysis, reaction recognition, tests and organic-pattern shortcuts under time pressure.

1. Atomic structure, periodic table and equations

ESAT SCOPE MARKERS

core Strictly C1-C17, with calculator-free calculation shortcuts.

trap A-level extras such as Hess cycles, K_c , ΔG , IR and electrochemical cells are intentionally removed.

SUBATOMIC PARTICLES

Particle	Relative mass	Charge	Location
proton	1	+1	nucleus
neutron	1	0	nucleus
electron	$\sim 1/1836$	-1	shell

ATOMIC AND MASS NUMBERS

Z = number of protons

A = protons + neutrons

neutrons = $A - Z$

Neutral atom: electrons = protons. Positive ion: fewer electrons.

Negative ion: more electrons.

ISOTOPES AND A_r

Isotopes have same proton number but different neutron number.

$A_r = \Sigma(\text{isotopic mass} \times \% \text{ abundance}) / 100$

trap Chemical behaviour is similar because electron arrangement is the same.

ELECTRON CONFIGURATION

First 20 elements fill shells as 2, 8, 8, 2 for ESAT purposes.

Group number predicts outer-shell electrons for main-group elements.

Period number predicts number of occupied shells.

COMMON ION CHARGES

shortcut Group 1 $\rightarrow +1$; Group 2 $\rightarrow +2$; Group 16 $\rightarrow -2$; Group 17 $\rightarrow -1$; aluminium $\rightarrow +3$.

Use roman numerals for variable metal ions, e.g. iron(III) = Fe^{3+} .

PERIODIC TABLE MAP

- Group 1: alkali metals.
- Group 2: alkaline earth metals.
- Group 16: common non-metals.
- Group 17: halogens.
- Group 18: noble gases.
- Transition metals: central block.

GROUP AND PERIOD TRENDS

Elements in same group have similar chemical properties because they have the same number of outer electrons.

Down a metal group, reactivity generally increases; down a non-metal group, reactivity generally decreases.

FORMULAE AND BALANCING

Reactions rearrange atoms/electrons; nuclei are not created or destroyed.

Balance atoms first, then charge for ionic equations.

State symbols: (s), (l), (g), (aq).

HALF-EQUATION BASICS

Oxidation loses electrons; reduction gains electrons.

Electrons appear on product side for oxidation and reactant side for reduction.

trap Both atoms and charge must balance.

2. Moles, concentration, gases and separation techniques

MOLES

$$n = m / M$$

$$m = nM$$

$$N = nN_A$$

Use formula mass in g mol^{-1} .

EMPIRICAL/MOLECULAR FORMULA

Empirical formula = simplest whole-number atom ratio.

Steps: mass $\rightarrow$ moles $\rightarrow$ divide by smallest $\rightarrow$ multiply to whole numbers.

$$\text{molecular multiplier} = M_r / \text{empirical formula mass}$$

EQUATION STOICHIOMETRY

Coefficients give mole ratios, not mass ratios.

shortcut Convert to moles $\rightarrow$ apply balanced ratio $\rightarrow$ convert to requested unit.

Limiting reagent gives the smaller possible product amount.

GAS VOLUME AT RTP

$$1 \text{ mol gas} \approx 24 \text{ dm}^3 = 24,000 \text{ cm}^3$$

$$n = V / 24 \text{ when } V \text{ is in dm}^3.$$

trap Gas-volume ratios follow mole ratios only at same T and P.

SOLUTIONS

$$c = n / V$$

$$n = cV$$

Use V in dm^3 for mol dm^{-3} .

$$1 \text{ dm}^3 = 1000 \text{ cm}^3$$

TITRATIONS

$$\text{moles} = \text{concentration} \times \text{volume}$$

Use balanced equation to convert acid moles $\leftrightarrow$ base moles.

trap Never assume 1:1 unless the equation says so.

YIELD AND PURITY

$$\% \text{ yield} = \text{actual yield} / \text{predicted yield} \times 100$$

$$\% \text{ purity} = \text{mass of pure substance} / \text{total sample mass} \times 100$$

SOLUBILITY AND SATURATED SOLUTIONS

Saturated solution contains maximum dissolved solute at that temperature.

Solubility usually read as mass dissolved per fixed solvent amount.

shortcut Use proportion scaling carefully when solvent amount changes.

SEPARATION TECHNIQUES

Mixture/problem	Technique
insoluble solid + liquid	filtration
soluble solid from solution	evaporation/crystallisation
liquids with different b.p.	simple/fractional distillation
immiscible liquids	separating funnel
coloured/soluble substances	chromatography
fine suspended solids	centrifugation

CHROMATOGRAPHY AND PURITY

$$R_f = \text{distance moved by spot} / \text{distance moved by solvent front}$$

Pure substance gives one spot; mixtures give multiple spots under same conditions.

3. Bonding, structure, energetics and rates

IONIC BONDING

Electrostatic attraction between oppositely charged ions.
High mp; conducts when molten/aqueous because ions move; brittle lattice.

METALLIC BONDING

Positive metal ions attracted to delocalised electrons.
Conducts electricity/heat; malleable because layers slide while bonding remains.

EXOTHERMIC/ENDOTHERMIC

Exothermic: releases heat; surroundings warm; products lower energy.
Endothermic: absorbs heat; surroundings cool; products higher energy.

shortcut On energy profiles, peak height above reactants = activation energy.

BOND ENERGY

$\Delta H \approx \Sigma \text{ bonds broken} - \Sigma \text{ bonds made}$
Breaking bonds absorbs energy; making bonds releases energy.
shortcut Count bonds after multiplying by equation coefficients.

COLLISION THEORY

- Higher concentration/pressure: more collisions per second.
- Higher temperature: particles move faster and more exceed activation energy.
- Smaller pieces/larger surface area: more exposed particles.
- Catalyst: alternative pathway with lower activation energy.

REVERSIBLE REACTIONS/EQUILIBRIUM

Dynamic equilibrium in closed system: forward rate = reverse rate; concentrations stay constant.
Increase concentration: shifts to use it up. Increase pressure: shifts to fewer gas moles. Increase temperature: shifts endothermic direction.
trap Catalyst speeds both directions equally; no equilibrium-position change.

COVALENT STRUCTURES

Small molecules: low mp/bp because weak intermolecular forces.
Giant covalent: very high mp because many strong covalent bonds.
trap Boiling simple molecules overcomes intermolecular forces, not covalent bonds.

INTERMOLECULAR FORCES

Stronger intermolecular forces → higher boiling point.
Larger molecules often have stronger London forces.
Hydrogen bonding needs H bonded to N, O or F.

CALORIMETRY

$q = mc\Delta T$
Use water as $c \approx 4.18 \text{ J g}^{-1} \text{ K}^{-1}$ when given/appropriate.
trap q is in J; convert to kJ for kJ mol^{-1} .

RATE OF REACTION

rate = change in amount/concentration/volume / time
Gradient of amount/concentration-time graph = rate.
rate $\approx 1 / \text{time}$ for fixed endpoint experiments.

4. Acids, bases, redox, electrolysis and qualitative tests

ACIDS AND BASES

Acid: H^+ donor / forms H^+ in water. Base: H^+ acceptor / forms OH^- if alkali.

Strong = fully ionised; weak = partially ionised. Concentrated = much solute; dilute = little solute.

COMMON ACID REACTIONS

- acid + metal $\rightarrow$ salt + hydrogen
- acid + carbonate $\rightarrow$ salt + carbon dioxide + water
- acid + hydroxide/oxide $\rightarrow$ salt + water
- non-metal oxides are generally acidic.

AGENTS AND DISPROPORTIONATION

Oxidising agent is reduced; reducing agent is oxidised.

Disproportionation: same element is both oxidised and reduced.

AQUEOUS PRODUCT SHORTCUTS

- At cathode: less reactive metal ions discharge; otherwise H_2 .
- At anode: halides give halogen; otherwise O_2 from OH^- /water.
- Electroplating: object to be plated is cathode.

GAS TESTS

Gas	Positive test
H_2	lit splint: squeaky pop
O_2	glowing splint relights
CO_2	limewater turns milky
Cl_2	damp blue litmus red then bleached white

FLAME TESTS AND WATER

Li^+ crimson; Na^+ yellow-orange; K^+ lilac; Ca^{2+} red-orange; Cu^{2+} green.

Water turns anhydrous $CuSO_4$ white $\rightarrow$ blue.

PH AND NEUTRALISATION

Lower pH means higher H^+ concentration; one pH unit = factor of 10.



Acid + base neutralisation is exothermic.

REDOX

Oxidation = oxygen gain, electron loss or oxidation-state increase.

Reduction = oxygen loss, electron gain or oxidation-state decrease.

shortcut OIL RIG: Oxidation Is Loss; Reduction Is Gain.

ELECTROLYSIS RULES

Cathode is negative: cations gain electrons/reduction.

Anode is positive: anions lose electrons/oxidation.

Molten ionic compound: metal at cathode, non-metal at anode.

Aqueous salts: water/ H^+ / OH^- may compete.

GROUP CHEMISTRY

- Group 1: reactivity increases down group; with water gives metal hydroxide + H_2 .
- Group 17: reactivity decreases down group; more reactive halogen displaces less reactive halide.
- Group 18: noble gases are unreactive because outer shell is full.

ION TESTS

Ion	Test/observation
CO_3^{2-}	acid gives CO_2 ; limewater milky
SO_4^{2-}	$BaCl_2$ + dilute HCl : white ppt
$Cl^-/Br^-/I^-$	$AgNO_3$ + dilute HNO_3 : white/cream/yellow
Cu^{2+}	$NaOH$: blue ppt
Fe^{2+}/Fe^{3+}	$NaOH$: green/brown ppt

5. Organic patterns, metals, particles, air and water

ORGANIC FAMILIES

Family	General formula / group
alkane	C_nH_{2n+2}
alkene	C_nH_{2n} , $C=C$
alcohol	$C_nH_{2n+1}OH$
carboxylic acid	$C_nH_{2n+1}COOH$
ester	-COO- link

COMBUSTION AND CRACKING

Complete combustion $\rightarrow CO_2 + H_2O$.

Incomplete combustion $\rightarrow CO$ and/or carbon soot plus water.

Cracking long-chain alkanes makes shorter alkanes + alkenes.

POLYMERS

Addition polymerisation: alkene $C=C$ opens; no small molecule lost.

Condensation polymerisation: monomers with two functional groups join and eliminate small molecules such as H_2O .

Polyester link: -COO-. Polyamide/protein link: -CONH-.

METALS

Reactivity relates to ease of forming positive ions.

More reactive metal displaces less reactive metal from compound.

Ores are often oxides; extraction usually involves reduction.

KINETIC PARTICLE THEORY

Solid: particles closely packed, vibrate. Liquid: close, move past each other. Gas: far apart, random rapid motion.

Gas pressure comes from particle collisions with container walls.

AIR, WATER AND POLLUTION

- Dry air is mostly N_2 and O_2 ; gases separated by fractional distillation.
- Greenhouse gases include CO_2 and CH_4 .
- Pollutants: CO , CO_2 , SO_2 , NO_x .
- Drinking water may contain chlorine and fluoride.

NAMING SHORTCUTS

Stems: meth-, eth-, prop-, but-, pent-, hex-.

Number chain to give functional group/branch the lowest possible locant.

Use di-/tri- for repeated groups.

ALKENES AND ADDITIONS

Alkenes decolourise bromine water.

- + H_2 with Ni catalyst $\rightarrow$ alkane.
- + $Br_2 \rightarrow$ dibromoalkane.
- + $HX \rightarrow$ haloalkane.
- + steam with acid catalyst $\rightarrow$ alcohol.

ALCOHOLS, ACIDS AND ESTERS

Alcohols react with sodium to form alkoxide + hydrogen.

Carboxylic acids are weak acids and react with carbonates to give CO_2 .

carboxylic acid + alcohol $\rightleftharpoons$ ester + water

USES AND TRANSITION METALS

Know broad uses/properties of Al, Fe, Cu, Ag, Au, Ti and alloys.

Transition metals often have variable oxidation states, coloured compounds and catalytic activity.

STATE CHANGES AND ENERGY

Melting/boiling require energy to overcome attractions; temperature stays constant during change of state.

Stronger forces/attractions usually mean higher melting/boiling point.